

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Fourth Semester of B.Tech.(CE/IT) Examination
December 2011

IT205 Operating System

Date: 08/12/2011

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 70

Instructions:

1. Figure to the right indicates full marks
2. Make suitable assumption whenever necessary and mention it clearly
3. Parts of same question should be answered together and in the same sequence.
4. Answers to the sections must be written separately

SECTION – I

- Q1**
- (a) How does distinction between monitor mode and user mode function as a simple form of protection (security) system? 4
 - (b) What is thread? How to implement threads in Kernel mode? 4
 - (c) Differentiate Deadlock prevention and Deadlock avoidance. 3
- Q2**
- (a) Explain Monolithic and Layered design structure of an Operating System. 6
 - (b) Explain Spin-lock in terms of busy waiting. 4
 - (c) Differentiate short-term and medium-term scheduler. 2
- OR**
- Q2**
- (a) Assuming three states of a process (Running, Ready, Blocked), draw and explain the state transition diagram. 6
 - (b) Write down components of PCB. 3
 - (c) What is the cause of thrashing? How does the system detect thrashing? Once it detects thrashing, what can the system do to eliminate this problem? 3
- Q3**
- (a) Disk requests came in to the disk driver for cylinders 10, 22, 25, 4, 38, 8, 3, 71 in that order. A seek takes 5Msec. per cylinder moved. How much seek time is needed for
 - i) FCFS algorithm
 - ii) Shortest-seek-time-first algorithm
 - iii) Elevator algorithm9
 - (b) Some systems support many types of structures for a file's data, while others simply support a stream of bytes. What are the advantages and disadvantages? 3
- OR**
- Q3**
- (a) Consider a swapping system in which memory consist of following hole sizes in memory cache. 9
10K , 40K , 20K , 18K , 7K , 9K , 12K , 15K
using following algorithms:
(1) First Fit (2) Best Fit (3) Worst Fit
Which hole is taken for successive segment request of (a) 12K (b) 10K (c) 9K

- (b) Explain directory structure in UNIX.

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SECTION – II

Q4

- (a) What is a page fault? Explain in detail. Under what circumstances do page faults occur? 4
- (b) Explain Encryption and Decryption with a suitable example. 4
- (c) What is deadlock? Explain recovery from deadlock by killing process. 3

Q5

- (a) Distinguish internal fragmentation and external fragmentation. Provide any two solutions to avoid external fragmentation. 5
- (b) Explain the functions of a line discipline for terminal driver. 5
- (c) Give the difference between exceptions and interrupt. 2

OR

Q5

- (a) Consider the following set of processes in order P1, P2, P3, P4 and P5 with the length of the CPU burst time given in milliseconds. Their priorities are 3, 5, 2, 1 and 4 respectively, with 5 being the highest priority, calculate average waiting time and turn-around time using given scheduling algorithm. 8

(1) SRTF

(2) RR (q=2)

(3) SJF

(4) Priority Scheduling (Preemptive)

Process	Arrival Time	Burst Time
P1	0	10
P2	1	9
P3	2	6
P4	3	7
P5	4	4

- (b) Explain Peterson's solution for avoiding race condition. 4

Q6 Answer the following questions (Any Four)

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- 1) Discuss about Windows (NT) file system in detail.
- 2) Explain about protection domain in UNIX.
- 3) Discuss the swapping in brief? How does buddy system speed up merging when process swaps out?
- 4) What are the basic functions of an operating system?
- 5) What is a semaphore? Which are operations done on semaphore?
- 6) Write a short note to illustrate resource allocation graph with deadlock condition.